TITLE PAGE

POCKETSMART AI: YOUR SMART BUDGET & RECOMMENDATION ASSISTANT

150-PAGE ACADEMIC PROJECT REPORT

Prerequisites and Full-Stack Development Project

Epic 1: Gemini AI Initialization | Epic 2: Core Functionalities Development | Epic 3: Backend – FastAPI Integration | Epic 4: UI Development | Epic 5: Testing and Deployment

CERTIFICATE

This project report titled “PocketSmart AI: Your Smart Budget & Recommendation Assistant” presents a structured academic study of prerequisites, cloud AI initialization, backend development, frontend development, testing, and deployment.

The report is organized into five development epics and a ten-page conclusion section.

DECLARATION

This project report is prepared for academic purposes. It describes the proposed design and implementation approach for PocketSmart AI and documents the technologies, architecture, testing practices, and deployment considerations used in the project.

ACKNOWLEDGEMENT

I sincerely acknowledge the support of faculty members, mentors, classmates, and technical documentation that helped in understanding Python, FastAPI, Google Cloud, Gemini AI, HTML, CSS, JavaScript, testing, and deployment.

ABSTRACT

PocketSmart AI is a smart budget and recommendation assistant that accepts financial inputs, processes budget information, and provides understandable recommendations. The application combines a web frontend, FastAPI backend, cloud-based Gemini AI, authentication routing, and structured testing.

This report covers prerequisites and five implementation epics. Epic 1 initializes Gemini AI and validates connectivity. Epic 2 develops core FastAPI functionality and login/register routing. Epic 3 establishes modular backend architecture, CORS, static routing, startup, and main functions. Epic 4 develops the HTML frontend. Epic 5 tests realistic budget inputs across domains and prepares deployment.

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Implementation, Security, Documentation and Future Scope

Conclusion – exactly 10 dedicated pages

PREREQUISITES – Project Background

This section establishes the knowledge, tools, environment, security practices, and development assumptions required before implementing PocketSmart AI. The goal is to make setup reproducible and reduce integration problems.

Project Background is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

PREREQUISITES – Problem Statement

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PREREQUISITES – Project Vision

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PREREQUISITES – Objectives

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PREREQUISITES – Scope

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PREREQUISITES – Functional Requirements

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PREREQUISITES – Non-Functional Requirements

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Non-Functional Requirements is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

PREREQUISITES – User Roles

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User Roles is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

PREREQUISITES – System Architecture

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System Architecture is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

PREREQUISITES – Technology Stack

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PREREQUISITES – Development Environment

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Development Environment is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

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Python Prerequisites is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

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Web Prerequisites is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

PREREQUISITES – API Prerequisites

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API Prerequisites is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

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PREREQUISITES – Security Prerequisites

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Security Prerequisites is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

PREREQUISITES – Documentation Prerequisites

This section establishes the knowledge, tools, environment, security practices, and development assumptions required before implementing PocketSmart AI. The goal is to make setup reproducible and reduce integration problems.

Documentation Prerequisites is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

PREREQUISITES – Prerequisites Checklist

This section establishes the knowledge, tools, environment, security practices, and development assumptions required before implementing PocketSmart AI. The goal is to make setup reproducible and reduce integration problems.

Prerequisites Checklist is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Epic Overview

This section establishes the Google Cloud and Gemini AI foundation. Configuration should follow current official Google Cloud and Gemini documentation, protect credentials, use appropriate permissions, and validate connectivity before application features depend on AI.

Epic Overview is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Google Cloud Account

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Google Cloud Account is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Cloud Project Creation

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Cloud Project Creation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – API Enablement

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API Enablement is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Gemini API Access

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Gemini API Access is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Credentials and Secrets

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EPIC 1 – GEMINI AI INITIALIZATION – Access and Permissions

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Access and Permissions is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – IAM Concepts

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EPIC 1 – GEMINI AI INITIALIZATION – Environment Configuration

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Environment Configuration is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – SDK Selection

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EPIC 1 – GEMINI AI INITIALIZATION – Prompt Design

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Prompt Design is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Budget Recommendation Prompt

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Budget Recommendation Prompt is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Structured Output

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Structured Output is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – AI Error Handling

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EPIC 1 – GEMINI AI INITIALIZATION – Connectivity Validation

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Connectivity Validation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Health Check

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Health Check is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Quota Awareness

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EPIC 1 – GEMINI AI INITIALIZATION – Cost Awareness

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Cost Awareness is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – AI Safety

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AI Safety is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Prompt Security

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EPIC 1 – GEMINI AI INITIALIZATION – Gemini Service Module

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Gemini Service Module is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Response Normalization

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EPIC 1 – GEMINI AI INITIALIZATION – Fallback Handling

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Fallback Handling is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 1 – GEMINI AI INITIALIZATION – Epic Validation

This section establishes the Google Cloud and Gemini AI foundation. Configuration should follow current official Google Cloud and Gemini documentation, protect credentials, use appropriate permissions, and validate connectivity before application features depend on AI.

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EPIC 1 – GEMINI AI INITIALIZATION – Epic Deliverables

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EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Epic Overview

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Epic Overview is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – FastAPI Installation

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

FastAPI Installation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Import Libraries

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Import Libraries is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Application Initialization

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Application Initialization is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Main Functionality Routing

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Main Functionality Routing is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Health Route

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Health Route is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Budget Input Route

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Budget Input Route is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Budget Calculation

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Budget Calculation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Recommendation Route

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Recommendation Route is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Authentication Overview

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Authentication Overview is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Registration Page Routing

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Registration Page Routing is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Login Page Routing

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Login Page Routing is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Authentication Validation

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Authentication Validation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Password Security

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Password Security is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Session or Token Concept

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Session or Token Concept is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Authentication Errors

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Authentication Errors is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Main Router Organization

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Main Router Organization is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Service Layer

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Service Layer is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Schema Layer

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Schema Layer is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Database Readiness

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Database Readiness is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Frontend API Calls

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Frontend API Calls is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Error Display

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Error Display is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Logging

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Logging is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Epic Testing

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Epic Testing is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 2 – CORE FUNCTIONALITIES DEVELOPMENT – Epic Deliverables

This section develops the core Python/FastAPI application. Routes, schemas, services, authentication, validation, and error handling are separated so the application remains testable and maintainable.

Epic Deliverables is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Epic Overview

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Epic Overview is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – FastAPI Route Design

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

FastAPI Route Design is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Router Modules

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Router Modules is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – APIRouter Usage

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

APIRouter Usage is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Versioned API

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Versioned API is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Modular Architecture Layers

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Modular Architecture Layers is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Core Configuration

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Core Configuration is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Service Boundaries

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Service Boundaries is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Schema Boundaries

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Schema Boundaries is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – CORS Purpose

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

CORS Purpose is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – CORS Configuration

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

CORS Configuration is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – CORS Credentials

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

CORS Credentials is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Static Routing Purpose

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Static Routing Purpose is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – StaticFiles Mount

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

StaticFiles Mount is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Frontend Hosting Choice

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Frontend Hosting Choice is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Startup Function

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Startup Function is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Lifespan Concept

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Lifespan Concept is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Main Function

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Main Function is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Uvicorn Startup

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Uvicorn Startup is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Dependency Injection

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Dependency Injection is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Exception Handling

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Exception Handling is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – API Documentation

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

API Documentation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Backend Testing

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Backend Testing is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Epic Validation

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Epic Validation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 3 – BACKEND – FASTAPI INTEGRATION – Epic Deliverables

This section organizes the FastAPI backend into modular routers, services, schemas, configuration, middleware, static resources, and startup logic. The objective is a clean API foundation that can grow without concentrating all logic in one file.

Epic Deliverables is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Epic Overview

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Epic Overview is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Frontend Architecture

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Frontend Architecture is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Landing Page

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Landing Page is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Registration Interface

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Registration Interface is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Login Interface

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Login Interface is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Dashboard Interface

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Dashboard Interface is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Budget Form

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Budget Form is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Expense Components

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Expense Components is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Recommendation Panel

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Recommendation Panel is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Responsive Design

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Responsive Design is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Accessibility

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Accessibility is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – API Integration

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

API Integration is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Loading State

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Loading State is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Error State

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Error State is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Success State

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Success State is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Modular JavaScript

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Modular JavaScript is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Modular CSS

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Modular CSS is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Static Assets

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Static Assets is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Frontend Security

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Frontend Security is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 4 – UI DEVELOPMENT – Epic Deliverables

This section develops the HTML frontend with CSS and JavaScript. The interface focuses on clear navigation, accessible forms, budget entry, recommendation display, responsive layout, and safe communication with backend APIs.

Epic Deliverables is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Epic Overview

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Epic Overview is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Testing Strategy

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Testing Strategy is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Test Environment

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Test Environment is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Student Budget Scenario

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Student Budget Scenario is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Office Employee Scenario

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Office Employee Scenario is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Freelancer Scenario

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Freelancer Scenario is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Family Budget Scenario

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Family Budget Scenario is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Small Business Scenario

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Small Business Scenario is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Input Validation Tests

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Input Validation Tests is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Authentication Tests

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Authentication Tests is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Gemini Connectivity Tests

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Gemini Connectivity Tests is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – API Integration Tests

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

API Integration Tests is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – CORS Tests

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

CORS Tests is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Static Routing Tests

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Static Routing Tests is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Performance Checks

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Performance Checks is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Security Testing

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Security Testing is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Deployment Preparation

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Deployment Preparation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Monitoring

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Monitoring is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Test Results Documentation

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Test Results Documentation is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

EPIC 5 – TESTING AND DEPLOYMENT – Epic Deliverables

This section validates the system with realistic, non-sensitive budget scenarios across different domains. It covers functional tests, API integration, authentication, CORS, AI connectivity, security, performance, deployment, and monitoring.

Epic Deliverables is treated as a separate project concern. The implementation should use clear naming, validated inputs, predictable outputs, reusable modules, and documentation. For PocketSmart AI, this concern connects the user interface, FastAPI services, and optional Gemini or database integrations. Development should distinguish configuration from source code and keep sensitive values outside the repository.

CONCLUSION 1 – OVERALL PROJECT SUMMARY

PocketSmart AI demonstrates an end-to-end software development workflow that combines web technology, Python backend services, cloud AI, authentication, testing, and deployment preparation.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 2 – PREREQUISITES

The prerequisites establish the required Python, web, API, cloud, Git, security, and documentation knowledge. A prepared environment creates a reliable starting point for implementation.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 3 – GEMINI AI INITIALIZATION

Epic 1 creates the cloud AI foundation by configuring Google Cloud, enabling the required Gemini service, managing access, protecting credentials, and validating connectivity.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 4 – CORE FUNCTIONALITIES

Epic 2 builds the central application behavior through FastAPI initialization, imports, main routes, budget processing, recommendation requests, and login/register routing.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 5 – FASTAPI BACKEND

Epic 3 demonstrates modular FastAPI architecture with routers, schemas, services, CORS, static routing, startup handling, and a clean application entry point.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 6 – USER INTERFACE

Epic 4 connects users to the system through HTML, CSS, and JavaScript. The UI supports authentication, budgeting, recommendation display, responsive design, and controlled error states.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 7 – TESTING AND DEPLOYMENT

Epic 5 validates the application using realistic budget scenarios and checks functional behavior, APIs, authentication, AI connectivity, CORS, security, and deployment readiness.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 8 – SECURITY AND RESPONSIBLE AI

The project treats financial information carefully. Credentials and passwords require protection, CORS should be restricted, HTTPS should be used in production, and AI recommendations should be presented as informational rather than professional financial advice.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 9 – FUTURE ENHANCEMENTS

Future versions can add persistent databases, richer analytics, notifications, multilingual support, mobile clients, stronger authentication, advanced recommendation models, monitoring, and automated deployment.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.

CONCLUSION 10 – FINAL CONCLUSION

The completed project provides a structured foundation for PocketSmart AI. Its five-epic architecture supports continued development while demonstrating practical skills in cloud AI, FastAPI, frontend development, testing, security, and deployment.

The project can be extended incrementally because responsibilities are separated into logical modules. Continued testing, documentation, security review, and user-focused refinement will support future versions.